Name

- For the ODE $y'' + 2y' + 26y = 0$
  - Write down the gen sol.

$y_c(t) = c_1 e^{-t} \sin(5t) + c_2 e^{-t} \cos(5t)$

  - Describe the solution as $t \to \infty$

As $t \to \infty$ wigling exp decay

$$\lambda^2 + 2\lambda + 26 = 0$$
$$(\lambda+1)^2 + 25 = 0$$
$$(\lambda+1)^2 = -25$$
$$\lambda + 1 = \pm 5i$$
$$\lambda = -1 \pm 5i$$

1. For the ODE $y'' \cancel{+} - 2y' + 26y = 0$

1.1. Write down the gen sol.

$y_c(t) = c_1 e^{t} \cos(5t) + c_2 e^{t} \sin(5t)$

1.2. Describe the solution as $t \to \infty$

As $t \to \infty$ exp growth and wiggles

$$\lambda^2 - 2\lambda + 26 = 0$$
$$(\lambda-1)^2 + 25 = 0$$
$$\lambda = 1 \pm 5i$$

$y_c(0) = c_1 e^{-4(0)} + c_2 e^{6(0)} = 10$

$\lambda^2 - 2\lambda - 24 = 0$

$\lambda = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$

$\lambda = -4$ and $\lambda = +6$

- For the ODE $y'' - 2y' - 24y = 0$ with $y(0) = 10$ and $y'(0) = 10$
  - Write down the solution

$c_1 = 5$
$c_2 = 5$

$y_c(t) = c_1 e^{-4t} + c_2 e^{+6t}$

  - Describe the solution as $t \to \infty$

As $t \to \infty$ $5e^{6t}$ big time exp growth

① $c_1 + c_2 = 10$
② $-4c_1 + 6c_2 = 10$

$4 \times ① + 2$

$10c_2 = 50$
$c_2 = 5$
$c_2 = 5$

2. For the ODE $y'' + 5y' + 6y = 0$ with $y(0) = 0$ and $y'(0) = 3$

2.1. Write down the solution

$y_c(t) =$

2.2. Describe the solution as $t \to \infty$

As $t \to \infty$